

Identifying Financially Constrained Firms: Data

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- Assume we can measure $Frictions_{i,t}$, what problem remains?
 - Basic endogeneity problem: correlation with firm productivity, management quality, etc.
 - This is a classic issue with triple diff specifications: the extra sorting variable technically should be also instrumented

Two broad approaches

1. Find “exogenous shifter” for $Frictions_{i,t}$
 - Δ collateral value, Δ working capital, Δ credit supply, access to foreign capital, Δ cash-flow
2. Time-series shocks interacted with ex-ante measures of financing frictions
 - Leverage, dividend payment, access to bond market, cash, 10-K, credit lines
3. Mix of both

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- *papers often don't report relevant economic elasticities but just “reduced form IV”*

- Very often we don't really care about how the exogenous shifter affects the LHS (e.g., “deregulation”)
- Same with shock like “bank shock:” in general people will report “one s.d. of RHS = x% of LHS,” but we don't really care
- It also makes it impossible to compare reduced form papers across them → Important for you to think about which greek letter you estimate!

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Brief overview of exogenous shifters: firm level

- Collateral
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- “Omnibus” bank credit supply shock + bond credit supply shock (Coppola, 2025)
 - Typical elasticity credit to K or Y ≈ 0.3 (0.26 [Cingano, Manaresi, Sette, 2016]; 0.36 [Amiti, Weinstein, 2018]; 0.23 [Pinardon-Touati, 2025]; 0.31 [Duquerooy, Matray, Saidi, 2025])
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- Cash-flow or cash-in-hand shocks
 - Multi-segment firms 0.12 [Lamont, 1997], pension shock? 0.6 [Rauh, 2006; Bakke, Whited, 2012], exchange rate 0.3 [Adams, Verdelhan, 2025]

Brief overview of exogenous shifters: “macro” level

- Bank deregulation across the US
 - First wave (Quincy, Xu, 25), second wave?, third wave?
- Collateral / bankruptcy reform: India (Vig, 13); Brazil (Fonseca, VonDoornik, 22); US (Favara et al., 21); Cross-countries (Haselmann, Pistor, Vig, 10)
- Foreign capital liberalization (Bau, Matray, 23)
- Financial shock transmission (Xu, 22)
- Bank or branch entry (Fonseca, Matray, 24; Quincy, Xu, 25)

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Identifying financing frictions: ex-ante sorting

1. Unless $Shock_{i,t}$ is aggregate, eq. is a triple-diff $\rightarrow$ likely wrongly specified
2. $Frictions_{i,t_0}$ potentially correlated with other stuff
 - $\rightarrow$ Need another instrument for $Frictions_{i,t_0}$ Hombert and Matray 2018

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3. Many proxies of financing frictions potentially measures the absence of financing frictions

- $Frictions_{i,t_0}$ potentially correlated with other stuff / proxy
 - e.g. firm productivity
- Proxies of financing frictions potentially measures the absence of financing frictions
 - Even if the proxy is correctly measure (say leverage) and NOT correlated with anything else (like productivity), it is unclear this measure FINANCING FRICTIONS
 - Potentially it can measure precisely the OPPOSITE, as we will discuss later $\rightarrow$ How can this be? ex-ante proxy is a EQUILIBRIUM OBJECT, should go in the other direction

Who is financially constrained?

1. High cash or low cash?
2. High leverage or low leverage?
3. With more undrawn credit lines or less undrawn credit lines?
4. Large (productive) exporting firms or small domestic firms?
5. Paying dividends or not paying dividends?

As cash increases, firms are less financially constrained

– Cash holding has increased a lot

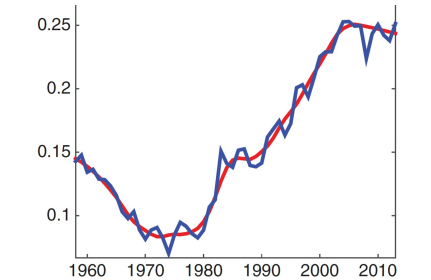


Fig. 1. Average cash-to-assets ratio of US-listed firms. This figure reports the average cash-to-assets ratio of US public companies over the period 1958-2013.

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– R&D firms are **less** financially constrained?

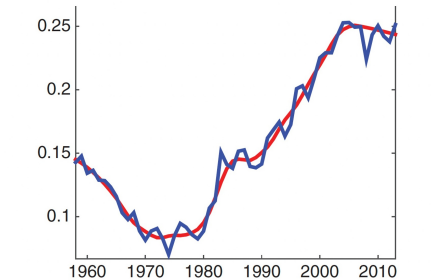


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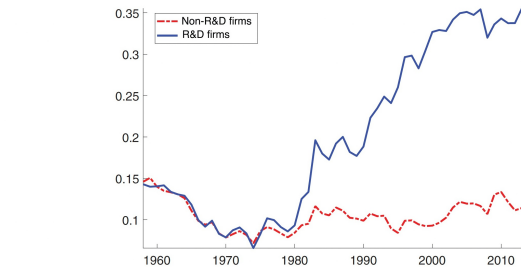


Fig. 3. Average cash-to-assets ratio of US listed firms by sector. This figure reports the average cash-to-assets ratio of R&D-intensive and non-R&D-intensive firms over the period 1958-2013. An R&D-intensive firm belongs to an industry (three-level digit SIC code) whose average R&D investment amounts to at least 2% of assets over the sample period.

As cash increases, firms are less financially constrained

→ Young R&D firms are **less** financially constrained?

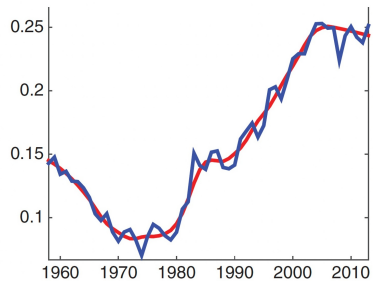
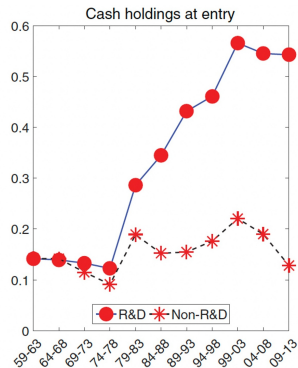


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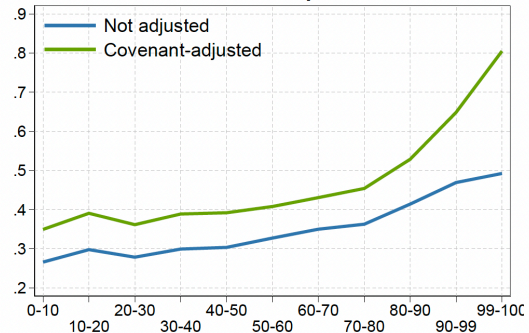
- Cash holdings rose for decades, yet the firms holding the most are young R&D firms.
- Ask the class: does high cash mean **unconstrained**, or precautionary saving by firms that cannot raise external finance cheaply?
- Plant the resolution: cash proxies **productivity and growth options**, not the absence of frictions.

Firms with more slack on their credit lines are less financially constrained

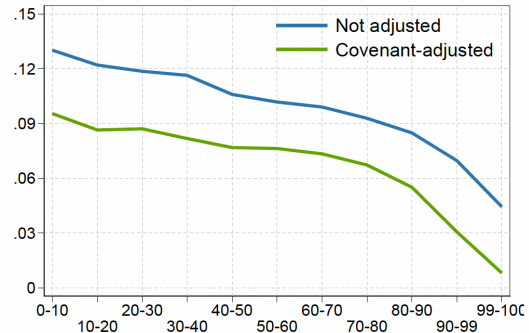
(Amberg, Jacobson, Quadrini, Picco (2023); Aydin, Kim (2026))

→ Large firms are **more** constrained?

C. Max utilization rate in past three months



D. Undrawn amount over assets

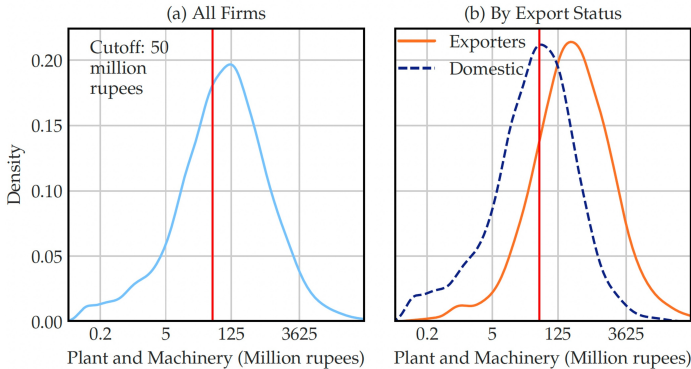


- More undrawn credit looks like slack, but large firms hold more lines and also look **more** constrained here.
- Ask: is an undrawn line spare capacity, or a precaution by firms that expect to be rationed?
- Same lesson as cash: the proxy tracks size and access, not the friction itself.

Exporters are more productive and bigger so should be less constrained

more productive bigger less (Finlay, 25)

Figure 1: Plant and Machinery Distribution



Exporters are more productive and bigger so should be less constrained

more productive
more

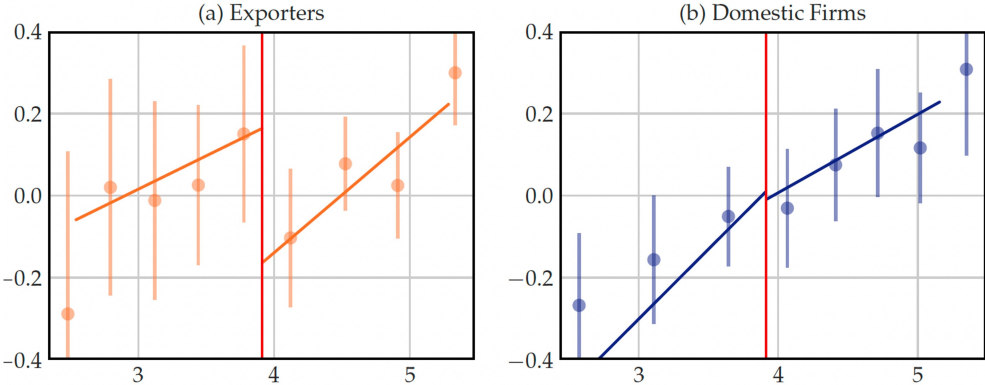
bigger

less

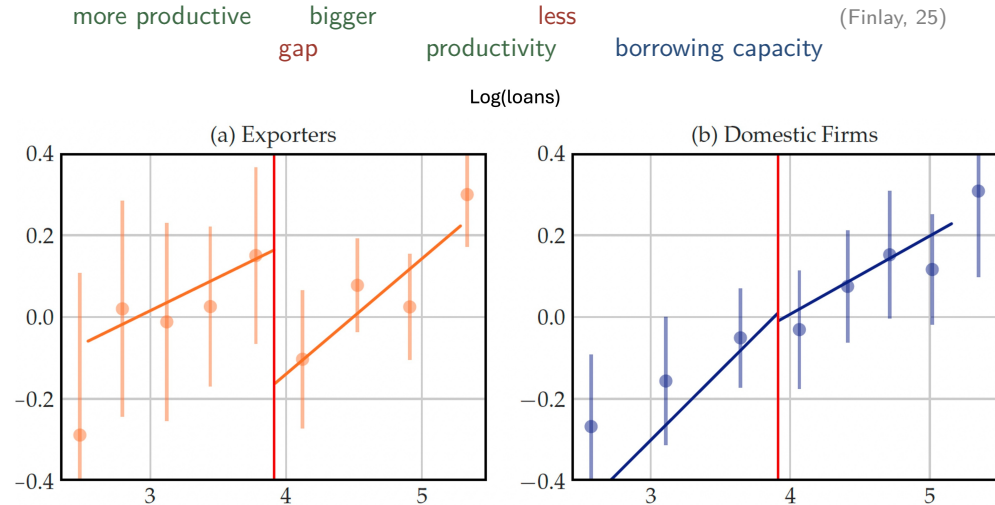
(Finlay, 25)

Xu, 2022; Finlay, 2025; Beaumont, Lenoir, 2025; Matray et al., 2025

Log(loans)



Exporters are more productive and bigger so should be less constrained



- *Financing frictions is the **gap** between **productivity** and **borrowing capacity***

- This is super important because it applies to a lot of context
- For instance: “Listed firms cannot be financially constrained because they are listed” → there is a reason they are listed, a priori they are more productive than private firms!
- Same argument for large firms. Whether large firms can be more or less financially constrained will depend on the development of financial market and whether financial markets are good at allocated resources to the most productive firms or not (Buera et al. 2012 AER). When financial markets are not developed, large firms are the ones with a lot of wealth but potentially limited productivity, while the constrained ones are productive.

Where does that leave us?

- Checking [what firms say](#) in their 10k: Hoberg-Maskimovic (2015)

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Corporate Debt Maturity and the Real Effects of the 2007 Credit Crisis. *Critical Finance Review*

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- Sorting firms on their [marginal revenue product of capital](#) (Bau-Matray, 2023; Matray-Mueller-Xu-Kabir, 2026)

1. *Checking what firms say in their 10k: Hoberg-Maskimovic (2015)*

- At least you are sure you capture something real

2. *Corporate Debt Maturity and the Real Effects of the 2007 Credit Crisis*

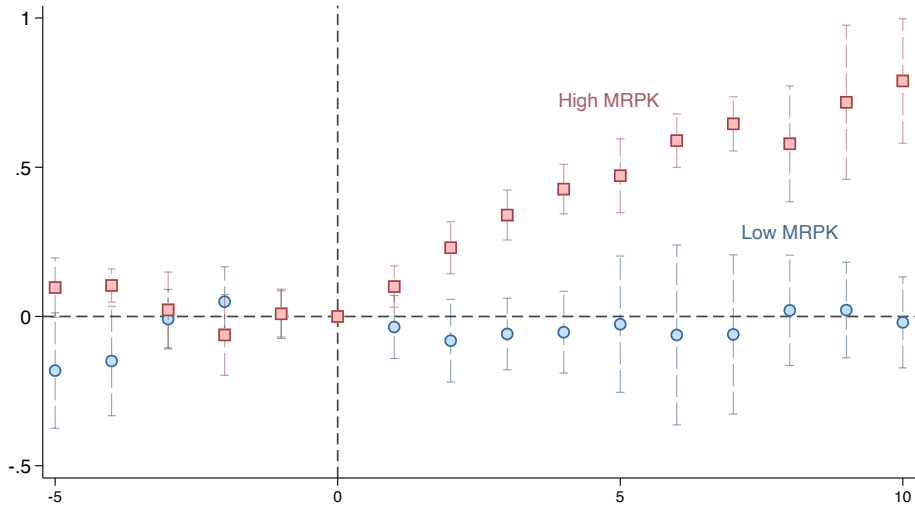
- At least this is a “within firm” measure so there is a sense you remove the firm heterogeneity
- Could not publish in top journals. Yet 1169 citations!!

Using MRPK to identify constrained firms

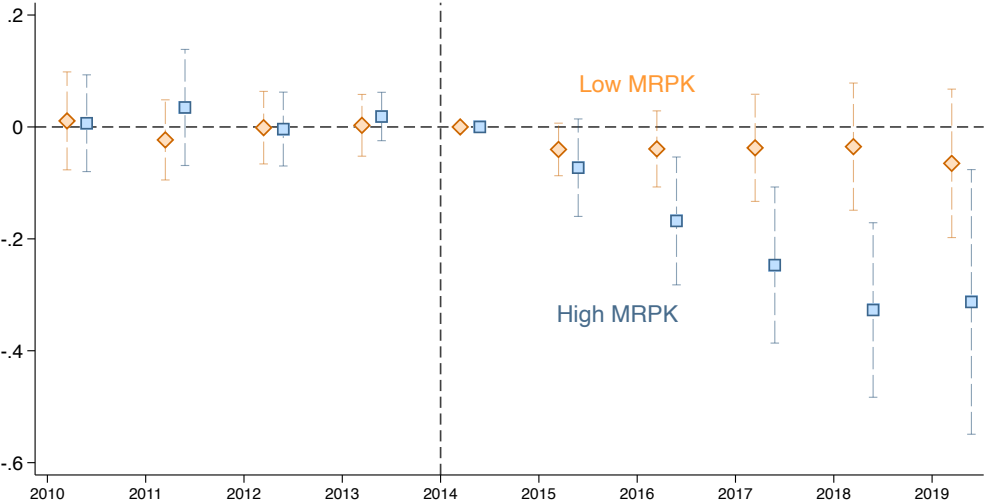
τ_i : reduced-form input-cost wedge = degree of financing frictions ($\tau_i = 0$ unconstrained, $\tau_i > 0$ constrained).

- Same user cost r across firms $\Rightarrow$ $MRPK_i$ ranks firms by their wedge τ_i
- High $MRPK \Rightarrow$ high $\tau \Rightarrow$ more constrained (capital too scarce at that firm)
- Dispersion in $MRPK$ = dispersion in financing frictions

MRPK example: Bau Matray (2023)



MRPK example: Matray Mueller Xu Kabir (2026)



What to take away

Key takeaway

Financing frictions are the **gap** between a firm's **productivity** and its **borrowing capacity**

- Most single balance-sheet proxies (cash, leverage, size, credit lines) can point the **wrong way** because they are **equilibrium objects** $\neq$ **primitives**

Key takeaway

Financing frictions are the **gap** between a firm's **productivity** and its **borrowing capacity**

- Most single balance-sheet proxies (cash, leverage, size, credit lines) can point the **wrong way** because they are **equilibrium objects** $\neq$ **primitives**
- Best solutions:
 - 10-K text, refinancing needs
 - **MRPK dispersion** $\rightarrow$ the only one with a clear **economic interpretation**

- State the one sentence aloud, then ask the class which of the five proxies from the “Who is constrained?” slide they would now trust.
- Remind them the gap framing is why listed or large firms are not automatically unconstrained.

Thank you!